

Ex. 38. What is the number of zeros at the end of the product of the numbers from 1 to 100?

Sol. Let $N = 1 \times 2 \times 3 \times \dots \times 100$.

Clearly, only the multiples of 2 and 5 yield zeros on multiplication.

In the given product, the highest power of 5 is much less than that compared to 2. So, we shall find the highest power of 5 in N .

$$\text{Highest power of 5 in } N = \left[\frac{100}{5} \right] + \left[\frac{100}{5^2} \right] = 20 + 4 = 24.$$

Hence, required number of zeros = 24.

Ex. 39. What is the number of zeros at the end of the product $5^5 \times 10^{10} \times 15^{15} \times \dots \times 125^{125}$?

Sol. Clearly, the highest power of 2 is less than that of 5 in N .

So, the highest power of 2 in N shall give us the number of zeros at the end of N .

$$\begin{aligned} \text{Highest power of 2} &= \text{Number of multiples of 2} + \text{Number of multiples of 4 (i.e. } 2^2) + \\ &\quad \text{Number of multiples of 8 (i.e. } 2^3) + \text{Number of multiples of 16 (i.e. } 2^4) \\ &= [(10 + 20 + 30 + \dots + 120) + (20 + 40 + 60 + \dots + 120) + (40 + 80 + 120) + 80] \\ &= (780 + 420 + 240 + 80) = 1520. \end{aligned}$$

Hence, required number of zeros = 1520.

Ex. 40. On dividing 15968 by a certain number, the quotient is 89 and the remainder is 37. Find the divisor.

$$\text{Sol. Divisor} = \frac{\text{Dividend} - \text{Remainder}}{\text{Quotient}} = \frac{15968 - 37}{89} = 179.$$

Ex. 41. A number when divided by 114, leaves remainder 21. If the same number is divided by 19, find the remainder. (S.S.C., 2010)

Sol. On dividing the given number by 114, let k be the quotient and 21 the remainder.

$$\text{Then, number} = 114k + 21 = 19 \times 6k + 19 + 2 = 19(6k + 1) + 2.$$

$\therefore$ The given number when divided by 19 gives remainder = 2.

Ex. 42. A number being successively divided by 3, 5 and 8 leaves remainders 1, 4 and 7 respectively. Find the respective remainders if the order of divisors be reversed.

$$\begin{array}{l} \text{Sol.} \quad \begin{array}{l|l} 3 & x \\ \hline 5 & y - 1 \\ \hline 8 & z - 4 \\ \hline & 1 - 7 \end{array} \quad \begin{array}{l} \therefore z = (8 \times 1 + 7) = 15; \\ y = (5z + 4) = (5 \times 15 + 4) = 79; \\ x = (3y + 1) = (3 \times 79 + 1) = 238. \end{array} \end{array}$$

$$\text{Now,} \quad \begin{array}{l|l} 8 & 238 \\ \hline 5 & 29 - 6 \\ \hline 3 & 5 - 4 \\ \hline & 1 - 2 \end{array}$$

$\therefore$ Respective remainders are 6, 4, 2.

Ex. 43. Three boys A, B, C were asked to divide a certain number by 1001 by the method of factors. They took the factors in the orders 13, 11, 7; 7, 11, 13 and 11, 7, 13 respectively. If the first boy obtained 3, 2, 1 as successive remainders, then find the successive remainders obtained by the other two boys B and C.

$$\begin{array}{l} \text{Sol.} \quad \begin{array}{l|l} 13 & x \\ \hline 11 & y - 3 \\ \hline 7 & z - 2 \\ \hline & 1 - 1 \end{array} \quad \begin{array}{l} \therefore z = 7 \times 1 + 1 = 8, \\ y = 11z + 2 = 11 \times 8 + 2 = 90; \\ x = 13y + 3 = 13 \times 90 + 3 = 1173. \end{array} \end{array}$$

$$\text{Now,} \quad \begin{array}{l|l} 7 & 1173 \\ \hline 11 & 167 - 4 \\ \hline 13 & 15 - 2 \\ \hline & 1 - 2 \end{array}$$

So, B obtained 4, 2 and 2 as successive remainders.

$$\text{And,} \quad \begin{array}{l|l} 11 & 1173 \\ \hline 7 & 106 - 7 \\ \hline 13 & 15 - 1 \\ \hline & 1 - 2 \end{array}$$

C obtained 7, 1 and 2 as successive remainders.